

TSMC Sustainability Direction Analysis

Project Objective: TSMC isn't just making chips, it's shaping the future of sustainability technology. Our project analyzes the TSMC 2024 Sustainability Report to uncover the language behind one of the world's most influential companies' ESG commitments. We aim to reveal hidden patterns, dominant themes, and the communication DNA across TSMC five sustainability roles. Most excitingly, we ask: Is TSMC's sustainability story driven by what they do or who they care for?

Source of Text: The TSMC 2024 Sustainability Report, sourced directly from TSMC's official document center, a rich, real-word corpus spanning climate action, talent development, supply chain responsibility, social impact, and green manufacturing across 270+ pages of authentic corporate ESG communication.

System Frameworks: Our pipeline transforms raw sustainability text into meaningful visual insight in four stages:

1. **Extraction:** Pull and segment text from the TSMC 2024 Sustainability Report.
1. **Processing:** Tokenization, stopword removal, TF-IDF scoring surface the most distinctive terms per role (not just the most frequent).
2. **Visualization:** Generate a categorized, role-specific word cloud dashboard with one tab per sustainability role.
3. **Interact:** One click reorganized everything: toggle between Action-Oriented and People-Centric perspective dynamically in the browser.

Expected Results: We're building more than a word cloud, we're building a lens to read corporate sustainability communication. Our deliverable includes:

1. A live, interactive web dashboard with tabbed word clouds per ESG role.
2. A click to toggle feature that instantly shifts between Action-Oriented and People-Centric views.
3. Keyword frequency charts and cross role heatmap map showing which themes bleed across multiple sustainability pillars.
4. An analytical insight into whether TSMC's language varies meaningfully by role; does an Admired Employer sound fundamentally different from an Innovation Pioneer?

This project lays the groundwork for Project 2, where we plan to extend toward sentiment analysis and potentially cross-year or cross-company ESG benchmarking.